

Control Estadístico de la Calidad

Examen Argumentativo

Nombre: _____ Matrícula: _____

Instrucciones: Muestra cómo obtienes tu respuesta. Puedes usar Excel o Minitab para tus cálculos de probabilidades o percentiles, pero no uses otros archivos o hagas uso de la IA. Respeta el código de ética del curso y del Tecnológico de Monterrey.

1. A Shewhart $\bar{X}$ chart is used with subgroup size $n = 6$. The process has $\bar{\bar{X}} = 75$ and $\bar{R} = 5.5$. Calculate the upper control limit (UCL) of the chart.
2. A process has specification limits $LSL = 106$ and $USL = 130$. The estimated process standard deviation is $\sigma = 4$. Calculate the potential capability index C_p .
3. A Shewhart control chart has center line 109, $UCL = 119$ and $LCL = 96$. A new sample statistic is 123.
What is the best conclusion?
 - (a) No action is needed; the point is within limits.
 - (b) The process variability has decreased.
 - (c) The process is out of statistical control.
 - (d) The process is centered and capable.
4. In an inspection of 100 units, 14 defective units were found. The fraction defective is $p = 0.14$. Which interpretation is correct?
 - (a) Approximately 14% of the inspected units were defective.
 - (b) Exactly 86% of future units will be conforming.
 - (c) The process capability index equals 0.14.
 - (d) The process is automatically in statistical control.
5. A process has $C_p = 1.9$ and $C_{pk} = 1.5$. Which conclusion is most appropriate?
 - (a) The process is out of statistical control.
 - (b) The process is perfectly centered between specifications.
 - (c) The specification limits are wider than the process spread because $C_{pk} > C_p$.
 - (d) The process spread is adequate, but the process is not perfectly centered.